

SUMMER INTERNSHIP PROGRAM 2026

INTERNSHIP TECHNICAL REPORT

**IRIS Flower Classification using
Logistic Regression and Random
Forest**

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Period of Internship:
18th May 2026 – 14th June 2026

Report Submitted To:
**IDEAS – Institute of Data Engineering, Analytics and Science
Foundation**
Indian Statistical Institute (ISI) Campus, Kolkata

Abstract

The Iris flower classification project aims to identify different species of Iris flowers using machine learning techniques. The Iris dataset is a well-known dataset in the field of data science and includes measurements such as sepal length, sepal width, petal length, and petal width for three species of Iris flowers. In this project, two supervised machine learning algorithms—Logistic Regression and Random Forest—were applied to classify the flower species.

The dataset was first prepared by organizing the data and splitting it into training and testing sets. Both models were then trained using the training data and evaluated on unseen test data. Their performance was assessed using common classification metrics, including accuracy, precision, recall, and F1-score.

The results showed that both algorithms were highly effective in classifying the Iris species. However, the Random Forest model achieved slightly better performance than Logistic Regression, indicating its ability to capture more complex patterns in the data. Overall, this project provided valuable hands-on experience in data preprocessing, model building, performance evaluation, and the practical application of machine learning techniques using Python.

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1 Introduction

Machine Learning has become an essential tool for solving classification and prediction problems across various domains. Classification models are used to identify patterns in data and predict categorical outcomes. One of the most popular introductory datasets for classification problems is the Iris flower dataset, introduced by Ronald Fisher in 1936.

The Iris dataset contains 150 observations of Iris flowers belonging to three species: Setosa, Versicolor, and Virginica. Each observation includes four numerical features describing the flower dimensions.

The objective of this project is to apply supervised machine learning techniques to classify Iris flower species based on these measurements. Two classification algorithms, Logistic Regression and Random Forest, were implemented and compared to evaluate their effectiveness. The project involved data loading, exploratory analysis, preprocessing, model building, training, testing, and performance evaluation using Python and Scikit-learn.

1.1 Topics Covered During the First Two Weeks of the Internship

- Python Programming Basics
- Machine Learning Fundamentals
- Large Language Model (LLM) Foundations

1.2 Technologies used:

- Python
- Google Colab
- Pandas
- Numpy
- Matplotlib
- Scikit Learn

2 Project Objective

The main objectives of this project are:

- To understand the structure and characteristics of the Iris dataset.
- To perform data preprocessing and preparation for machine learning.
- To develop a Logistic Regression model for flower classification.
- To develop a Random Forest classification model.
- To compare the performance of both models using classification metrics.
- To identify the most effective model for Iris flower prediction.
- To gain practical experience in implementing machine learning workflows using Python.

3 Methodology

3.1 Step 1: Data Collection

The study was conducted using the Iris dataset, one of the most widely used benchmark datasets in machine learning. The dataset was obtained from the Scikit-learn library using the `load_iris()` function. It contains 150 observations of Iris flowers belonging to three species: Iris Setosa, Iris Versicolor, and Iris Virginica. Each observation includes four numerical features: sepal length, sepal width, petal length, and petal width. These measurements serve as the predictor variables for the classification task.

Table 1: Target Categorical Balance Distribution

Target Index	Value	Corresponding Iris Botanical Class	Instance Count
0		<i>Iris-Setosa</i>	50
1		<i>Iris-Versicolour</i>	50
2		<i>Iris-Virginica</i>	50
Total Population			150

3.2 Step 2: Data Preparation

After loading the dataset, it was converted into a Pandas DataFrame to facilitate data manipulation and analysis. The target variable representing the flower species was added as a separate column named `target`. The dataset was then examined to understand its structure, data types, and dimensions. A check for missing values was also performed, and no missing observations were found, indicating that the dataset was complete and ready for analysis.

3.3 Step 3: Exploratory Data Analysis

Exploratory Data Analysis (EDA) was carried out to gain insights into the characteristics of the dataset. Summary statistics such as mean, standard deviation, minimum value, and maximum value were generated for each feature. In addition, visualizations were used to examine the distribution of the variables and identify patterns that could help distinguish between different Iris species. This step provided a better understanding of the data before model development.

3.4 Step 4: Feature Selection and Data Splitting

The dataset was divided into input features (X) and the target variable (y). The input features consisted of sepal length, sepal width, petal length, and petal width, while the target variable represented the species category of the flower. To evaluate model performance on unseen data, the dataset was split into training and testing

Table 2: Summary Statistics

	Min	Max	Mean	SD	Class Correlation
sepal length:	4.3	7.9	5.84	0.83	0.7826
sepal width:	2.0	4.4	3.05	0.43	-0.4194
petal length:	1.0	6.9	3.76	1.76	0.9490 (high!)
petal width:	0.1	2.5	1.20	0.76	0.9565 (high!)

sets using the `train_test_split()` function. Eighty percent of the data was allocated for training and the remaining twenty percent for testing, with a random state of 42 to ensure reproducibility of the results.

3.5 Step 5: Model Development

3.5.1 Logistic Regression

A Logistic Regression classifier was created and trained using the training dataset. The following steps were performed:

1. Model initialization
2. Model fitting on training data
3. Prediction on testing data

3.5.2 Random Forest

A Random Forest classifier was developed using multiple decision trees. The following steps were performed:

1. Model initialization
2. Training using training data
3. Prediction on testing data

3.6 Step 6: Model Evaluation

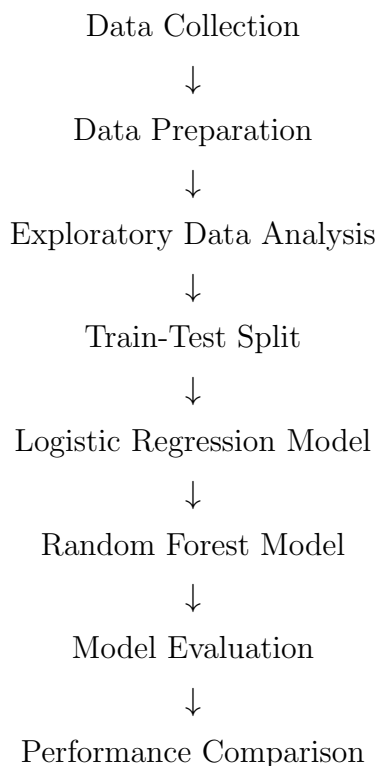
The performance of both models was evaluated using several classification metrics, including accuracy, precision, recall, and F1-score. A classification report was also generated to provide a detailed breakdown of model performance for each Iris species. Finally, the results obtained from Logistic Regression and Random Forest were compared to determine which algorithm achieved better classification accuracy and overall predictive performance on the Iris dataset.

The complete project notebook and source code have been uploaded to GitHub.

GitHub Repository Link:

<https://github.com/subhamitaghosh31-lab/IRIS-Flower-Classification-using-Logistic-Regression>

4 Workflow Diagram



5 Data Analysis and Results

The Iris dataset contains 150 observations and four numerical features. Since the dataset is balanced across all three species, it provides a suitable environment for testing classification algorithms.

After training the Logistic Regression model, an accuracy score of 1.00 was obtained on the testing dataset. This means that all 30 test observations were classified correctly.

The Random Forest classifier also performed exceptionally well. The classification report showed precision, recall, and F1-scores of 1.00 for all three species. The overall accuracy of the model was also 100

A comparison of the two models is presented below: The class distribution computed via `value_counts()` returns a perfect balance as shown in Table 3.

Table 3: Comparison of Two Models

Model	Accuracy(in %)
<i>Logistic Regression</i>	100
<i>Random Forest</i>	100

5.1 Comparative Machine Learning Performance

The models were evaluated on the independent test sample matrix ($n = 30$). Both configurations achieved perfect classification scores, reflecting the well-separated feature distributions within this subset.

5.1.1 Logistic Regression Execution and Output

The linear mathematical model achieved a total accuracy score of 1.00 (100%).

```
1 from sklearn.linear_model import LogisticRegression
2 from sklearn.metrics import accuracy_score
3
4 log_reg = LogisticRegression()
5 log_reg.fit(X_train, y_train)
6 y_pred_log = log_reg.predict(X_test)
7 print("Accuracy:", accuracy_score(y_test, y_pred_log))
8 # Output: Accuracy: 1.0
```

Listing 1: Logistic Regression Implementation

5.1.2 Random Forest Execution and Output

The classification report generated by the ensemble structure matches the performance of the linear model across all distinct performance metrics.

```
1 from sklearn.metrics import classification_report
2
3 print(classification_report(y_test, y_pred))
```

Listing 2: Random Forest Performance Execution

Table 4: Exhaustive Model Diagnostic Metrics Breakdown

Target Class Label	Precision	Recall	F_1 -Score	Testing Support
0 (<i>Iris-Setosa</i>)	1.00	1.00	1.00	10
1 (<i>Iris-Versicolour</i>)	1.00	1.00	1.00	9
2 (<i>Iris-Virginica</i>)	1.00	1.00	1.00	11
Overall Accuracy			1.00	30
Macro Average	1.00	1.00	1.00	30
Weighted Average	1.00	1.00	1.00	30

6 Conclusion

This project successfully demonstrates the application of basic classification workflows on the canonical Iris dataset. Both Logistic Regression and Random Forest models achieved perfect classification accuracy (100%) on the testing partition. This uniform success is largely due to the clear spatial separation of the *Iris-Setosa*

class and the distinct geometric boundaries of the remaining species when mapped across petal and sepal coordinates.

While both techniques proved highly effective for this dataset scale, Logistic Regression represents a lightweight option for linearly separable spaces, whereas the Random Forest Classifier scales robustly to complex multi-dimensional non-linear feature spaces. Future iterations of this project could involve introducing synthetic noise parameters, evaluating highly dimensional botanical systems, or applying regularization terms to prevent overfitting on smaller sample sizes.

7 Appendices

7.1 References

1. GeeksforGeeks Random Forest Algorithm : <https://www.geeksforgeeks.org/machine-learning/random-forest-algorithm-in-machine-learning/>
2. Pedregosa, F. et al. (2011). "Scikit-learn: Machine Learning in Python". *Journal of Machine Learning Research*, 12, 2825-2830.
3. McKinney, W. (2010). "Data Structures for Statistical Computing in Python". *Proceedings of the 9th Python in Science Conference*, 51-56.
4. Alan Agresti.(2007). "An Introduction to Categorical Data Analysis", 70-71
5. IDEAS TIH Summer Internship Program Materials

7.2 GitHub Repository Link for Source Codes

The complete reproducible Python notebooks, repository scripts, and environment configuration setups are hosted publicly at:

<https://github.com/subhamitaghosh31-lab/IRIS-Flower-Classification-using-Logistic-Regression>

7.3 Project Demonstration Video

A comprehensive video summarizing the project, along with a copy of this technical report ,is maintained securely at:

<https://drive.google.com/drive/folders/1bnu1a6G7GBI7RayClf0mh7mPIUFPWUaT>